Acceleration of Quantum Kernel Methods for Image Classification Tasks

Project Work Parallel Programming

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Overview

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2. Experimental Methodology
3. HPC Architecture
4. Performance Benchmark
5. Classification Results
6. Failure Analysis: Kernel Collapse
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Context & Scope

Context & Motivation

Quantum Kernel SVMs

- Project classical data into a 2^Q-dimensional Hilbert space.
- Fidelity kernel: $k(x_i, x_j) = |\langle\psi(x_i)|\psi(x_j)\rangle|^2$.
- At $Q = 16$: 65,536 complex amplitudes per state.
- Guaranteed convex SVM optimization.

HPC Bottleneck

- Gram matrix cost: $\mathcal{O}(N^2 \cdot 2^Q)$.
- GPU/CUDA acceleration is essential at scale.

Three Research Questions

1. How do classical and quantum-kernel SVMs compare on the same binary tasks?
2. Which backend offers the best throughput–runtime trade-off at 16 qubits?
3. Which empirical signals indicate kernel concentration?

Experimental Methodology

Datasets & Binary Tasks

Three datasets, binarized into three difficulty levels:

Dataset	Easy	Med	Hard
Fashion-MNIST	0 vs 1	3 vs 8	0 vs 6
CIFAR-10	1 vs 6	0 vs 2	3 vs 5
SVHN	1 vs 0	2 vs 7	3 vs 5

Default Setup

- $Q = 16$ qubits $\rightarrow 2^{16}$ amplitudes
- **YZ ansatz** with entangling layers (fixed weights)
- 3-fold CV + Optuna for hyperparameter C
- Metrics: F1, AUC, runtime, SV fraction, kernel std

Preprocessing Pipeline & Fidelity Kernel

Feature pipeline:

1. Extract pixels and flatten
2. PCA to $d = 16$ components
3. Range scaling
4. Encode via *angle map*, *Y map* or *YZ map*
5. Simulate states via PennyLane

Fidelity kernel:

$$K_{ij} = \left| \sum_{k=1}^{2^Q} S_A[i,k]\, \overline{S_B[j,k]} \right|^2$$

Training protocol:

- Train/test split from dataset
- 3-fold CV on training partition
- Hyperparameter search (C) via Optuna
- Final re-fit on train+val, eval on test set

Kernel-based, not variational

Weights are fixed at init – kernel method, not a training loop.

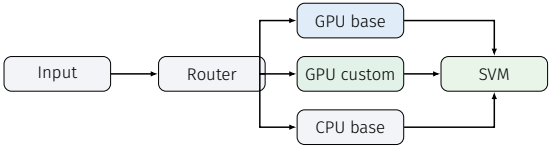
HPC Architecture

Unified Backend API

Three execution paths, same normalization and SVM:

- **CPU base** – NumPy reference, no VRAM tiling
- **GPU base** – PyTorch GPU matmul + pinned memory + CUDA streams
- **GPU custom** – PyTorch states → DLPack → CuPy → custom CUDA kernel (FP64)

SVM solver always runs on CPU.



Backend Optimizations & Shared Normalization

Key optimizations:

- **VRAM-aware tiling**: block size auto-tuned to budget
- **Kernel autotuning**: tile sizes cached persistently
- **Bulk precompute**: states prebuilt when memory allows
- **Stream pool**: CUDA stream orchestration
- Best: vram_fraction=0.95, num_streams=1, GPU base pinned+streams

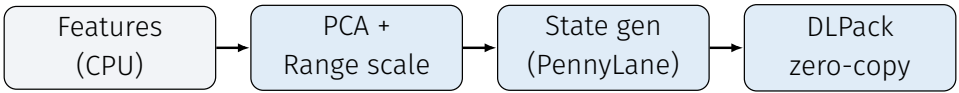
⚖ Shared Normalization

All backends use identical post-processing:

$$\hat{K}_{ij} = \frac{K(X,Y)_{ij}}{\sqrt{K(X,X)_{ii}\,K(Y,Y)_{jj}}}$$

- NaN/Inf checks before writing
- Diagonal validation before SVM fit
- Guarantees fair comparison

GPU Custom: Kernel Construction Pipeline



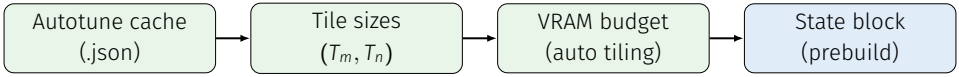
Host preprocessing (step 1/3):

- Raw image pixels extracted and flattened
- PCA to 16 components + range scaling
- Quantum states generated via PennyLane in **FP64**
- Output: complex state matrix $S \in \mathbb{C}^{N \times 2^Q}$

Why FP64?

FP32 leads to **kernel collapse** ($AUC \approx 0.5$) due to floating-point cancellation in the fidelity dot products at 2^{16} dimensions.

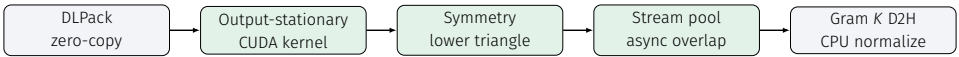
GPU Custom: Control & Cache Layer (2/3)



How the control layer works:

- **Autotune cache**: tile sizes (T_m, T_n) measured once per (nq, dtype) and saved to JSON
- **VRAM-aware tiling**: vram_fraction=0.95 caps state block size to available GPU memory
- **Bulk precompute**: if the full block fits, states are prebuilt to reduce PennyLane round-trips
- Optimal profile in benchmarks: **1 stream** with **pinned memory** (GPU base path)

GPU Custom: Device Core (3/3)



Device execution:

- **DLPack**: PyTorch GPU tensor $\rightarrow$ CuPy with **zero memory copy**
- **Output-stationary**: each CUDA thread writes exactly one K_{ij} (no write conflicts)
- **Symmetry**: lower triangle only; then $K_{ji} \leftarrow K_{ij}$ mirrors upper entries
- **Stream pool**: asynchronous block overlap between state prep and Gram compute
- Final step: D2H transfer of K and CPU-side cross-normalization

Performance Benchmark

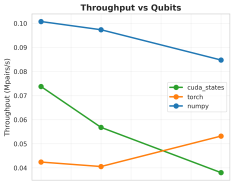
Throughput by Dataset & Backend (Mpairs/s)

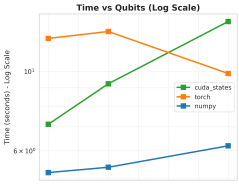
Highest qubit setting per dataset (Fashion: 12 qubits; CIFAR-10/SVHN: 16 qubits)

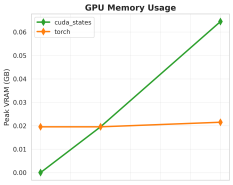
Dataset	Size	Qubits	CPU base	GPU base	GPU custom
fashion	256	12	0.0059	0.0140	**0.0159**
fashion	512	12	0.0224	**0.0299**	0.0298
fashion	1024	12	**0.0856**	0.0575	0.0546
cifar10	256	16	0.0020	0.0057	**0.0062**
cifar10	512	16	0.0086	**0.0175**	0.0112
cifar10	1024	16	0.0319	**0.0375**	0.0245
svhn	256	16	0.0023	0.0053	**0.0058**
svhn	512	16	0.0097	**0.0175**	0.0124
svhn	1024	16	0.0328	**0.0366**	0.0243

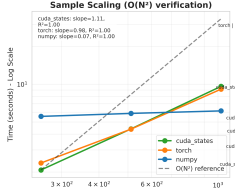
At size 256: GPU backends dominate. At size 512: GPU base fastest. At size 1024: CPU base recovers on Fashion.

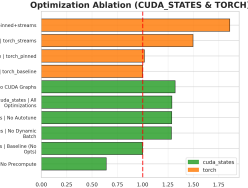
Benchmark Profile – Fashion-MNIST

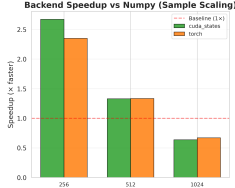












Backend ranking is **not global**: it varies with sample size and dataset profile.

Benchmark Profiles – CIFAR-10 & SVHN

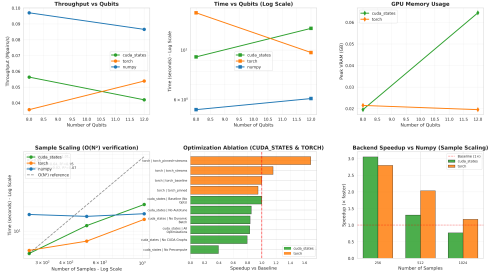


Figure 1: CIFAR-10 (16 qubits)

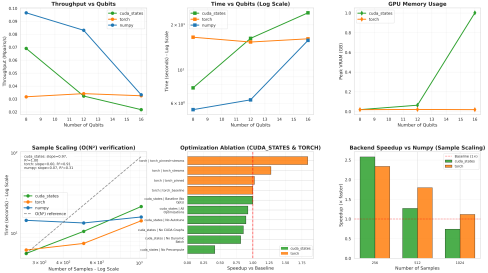


Figure 2: SVHN (16 qubits)

GPU base is fastest across most 16-qubit configurations. Data movement dominates over stream complexity.

Classification Results

Quantum Backend Quality vs Runtime

Mean F1, AUC and runtime – averaged over difficulty levels, latest run extraction

Dataset	Size	Backend	Mean F1	Mean AUC	Mean time (s)
cifar10	500	GPU custom	0.3500	0.4480	336.13
cifar10	500	GPU base	0.4270	0.4812	190.61
cifar10	1000	GPU custom	0.4305	0.4365	496.73
cifar10	1000	GPU base	0.5729	0.5532	302.19
fashion	500	GPU custom	0.8905	0.9569	237.05
fashion	500	**GPU base**	**0.8889**	**0.9574**	**110.78**
fashion	1000	GPU custom	0.8958	0.9553	339.29
fashion	1000	**GPU base**	**0.8998**	**0.9557**	**187.13**
svhn	500	GPU custom	0.1789	0.4719	545.39
svhn	500	GPU base	0.1547	0.4719	302.73
svhn	1000	GPU custom	0.1596	0.4846	637.30
svhn	1000	GPU base	0.2998	0.4846	380.61

GPU base generally faster and more accurate. Fashion strong; SVHN/CIFAR-10 degrade.

Classical vs Quantum: Paired Comparison

Positive ΔF1/ΔAUC = classical better; negative ΔTime = classical faster

Comparison	Size	ΔF1	ΔAUC	ΔTime (s)	Wins C/T/Q
Classical — GPU base	500	+0.1708	+0.1488	−188.28	8/0/1
Classical — GPU base	1000	+0.1721	+0.1600	−274.52	8/1/0
Classical — GPU custom	500	+0.1879	+0.1600	−359.76	8/0/1
Classical — GPU custom	1000	+0.2676	+0.1990	−475.66	8/1/0

- Classical **dominates**: wins 8/9 paired comparisons at both sample sizes
- Quality gap **widens at 1000 samples** – quantum kernels do not scale better with data
- Acceleration improves throughput but does **not remove representational limits**

Failure Analysis: Kernel Collapse

Hard Tasks: Quantitative Failure Pattern

Observable collapse signals:

- Inter-class gap ≈ 0 after normalization
- Kernel std $\in [10^{-4}, 10^{-3}]$
- SV fraction $\rightarrow 1.0$
- AUC ≈ 0.5 (random)

Hard CIFAR-10 with 6 layers:

- *kernel std = 0.0005*
- *SV frac = 1.0000*
- *test F1 = 0.0000*
- *test AUC = 0.4757*

⃠ Kernel Concentration

States become nearly **indistinguishable**. All samples become support vectors – the SVM cannot separate classes.

Contributing factors:

- PCA/grayscale information loss
- Generic ansatz, not image-aware
- Deeper circuits $\rightarrow$ more concentration
- Global fidelity kernel limitation

Difficulty-Level Summary Across Backends

Mean F1, AUC and time per difficulty (averaged over datasets and sizes, latest run)

Difficulty / Backend	F1	AUC	Time (s)
easy / GPU custom	0.4907	0.6327	475.44
easy / GPU base	0.6366	0.6499	258.76
med / GPU custom	0.5263	0.6314	404.06
med / GPU base	0.6560	0.6896	226.39
hard / GPU custom	0.4357	0.6125	416.44
hard / GPU base	0.3289	0.6126	251.88

- **GPU base** outperforms at easy/med – faster and more accurate
- At **hard**, both backends collapse: F1 drops, AUC $\rightarrow$ 0.5
- Compute acceleration cannot resolve the representational bottleneck

Conclusion & Perspectives

Conclusion

- **HPC Pipeline:** Three backends share a unified API. GPU paths dominate small sizes; ranking varies with dataset and sample count.
- **Fashion quality:** Both quantum backends competitive (F1 up to **0.9653**, AUC up to **0.9956** for GPU base, size 1000).
- **Classical dominates:** RBF SVM wins 8/9 paired comparisons at both sample sizes. Gap widens at 1000 samples.
- **Kernel collapse on hard tasks:** Concentration signals (std $\sim 10^{-4}$, SV frac = 1). Acceleration alone does not fix this.

Perspectives

→ Better Feature Maps

- Projected/local kernels to reduce concentration
- Geometry-aware ansatz for image structure
- Stronger classical front-end before quantum embedding

→ Richer Diagnostics

- Gap trajectories and kernel spectra
- SV dynamics to detect collapse earlier
- Broader benchmark grids per dataset

→ HPC Scaling

- Multi-GPU / MPI for large Gram builds
- GPU-side SVM solvers to remove CPU bottleneck

Thank You!

github.com/DylanUnifi

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Questions?